HITACHI

SERVICE BULLETIN

CONFIDENTIAL

DESCRIPTION

SERVICE BULLETIN CODE
3 2011 0110

FILING LOCATION
ZX-3-All-269
ZX-5-All-202

SUBJECT

AIR CONDITIONER REFRIGERANT INSPECTION PROCEDURE AND FAILURE CONFIRMATION METHOD BY USING PRESSURE GAUGE

OUTLINE

The refrigerant inspection window (sight glass) has been abolished for ZX-3 and onward. This Service Bulletin explains the simple inspection method and investigation method to probe the cause of faulty cooling.

Thoroughly read the safety section in the service manual before starting the work to ensure safe operation. Welding work, if any, should only be done by qualified welders.

APPLICABLE MODELS / SERVICE BULLETIN NO. FOR EACH MODEL

Machine Model(s): ZX-3, ZX-5

Machine Serial Nos.:

DESCRIPTION

Refer to the following page(s).

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1 Conduct the regular inspection by following the procedure stated below

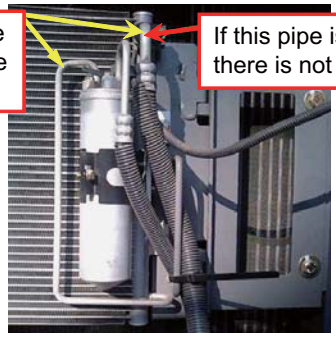
① Set the temperature to minimum (FC). Turn up the air volume to MAX and operate the engine for 2-3 minutes at high idle.

② Check to see if cold air ows out from the indoor vent.

③ If cold air is not coming out, check the temperature difference (about 20℃) between the condenser intake pipe and outtake pipe. (Con rm if the surface of intake pipe – thicker pipe – is hot (about 70℃ or higher).)
※ See the photo below.

④ -1. If the intake pipe of the condenser is not hot (about 70℃), it means that there is not enough refrigerant and the temperature difference with the outtake pipe (receiver tank side) is clearly small (it is not hot to the touch).
・In case of refrigerant shortage, don't charge. Instead, collect the refrigerant inside a refrigerant collector. Follow the charging instructions to charge the refrigerant and check for leaks.

④ -2. If the above pipe is hot, check if condenser, evaporator, etc. are blocked.



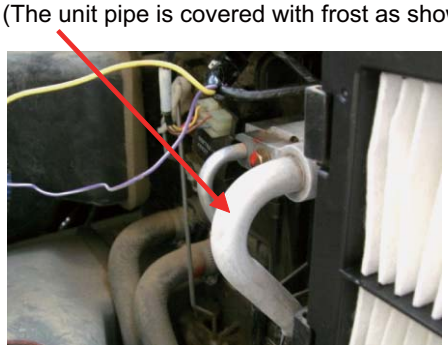
2 Failure confirmation procedure

2-1 Faulty operation phenomenon of compressor relay (stuck contact point, etc.)

1) While being used, the indoor air volume drops. Water fails to drain from the drain hose. (Check after air volume has dropped.)
2) When phenomenon 1) is detected, turn off the air conditioner. Leave for about one hour and then, turn back on. Then, air will ow out once again from vent.

※ If the air chills only when machine is restarted (UP air outtake), it means that the air conditioner is in freezing phenomenon.

(The unit pipe is covered with frost as shown in the attached photo.)



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Countermeasures
① Replace the compressor relay.
② There is the possibility of damage. Replace the compressor and dryer.

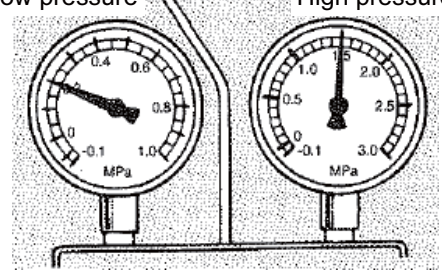
※ If internal damage of compressor is con rmed, replace all parts, including the refrigerating hose.

2-2 Confirmation by using pressure gauge (example when the temperature is 30℃ to 35℃)

◎ Measurement conditions = Warm up the engine.

- Cab door: Open
- Temperature setting: Maximum cool
- Air conditioner: ON
- Intake air suction mode
- Air volume: MAX
- Engine rotation number: MAX (no load)

Normal state

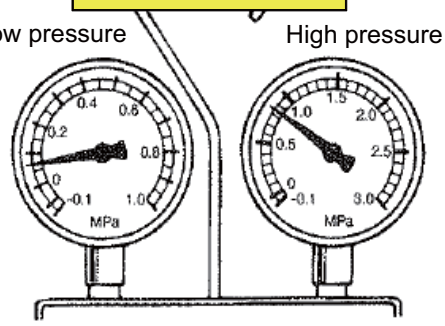


The fig. at left shows a general case.

Low pressure side: 0.15 - 0.25 MPa
High pressure side: 1.37 - 1.57 MPa

The gauge display may change somewhat depending on the conditions.

Insufficient refrigerant

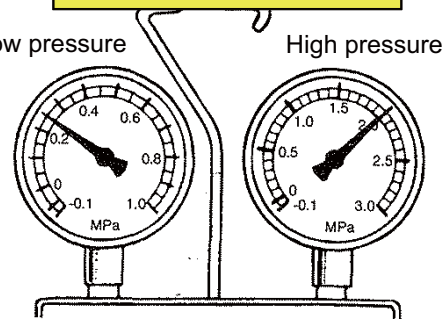


● Condition
- Doesn't cool properly
- Pressure reading on both the low and high pressure sides are low.

● Remedy
- Release the refrigerant and charge as needed.
- Check for leak.

Too much refrigerant

Faulty chilling on condenser



● Condition
- Doesn't cool properly.
- Pressure reading on both the low and high pressure sides are high.

● Remedy
- Release the refrigerant and charge as needed.
- Clean the condenser.

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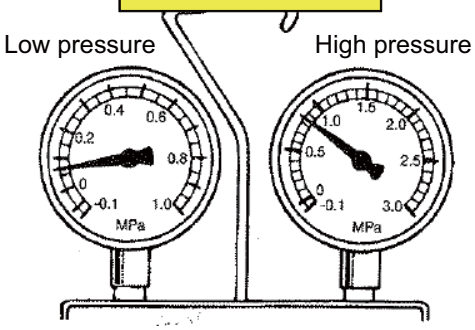
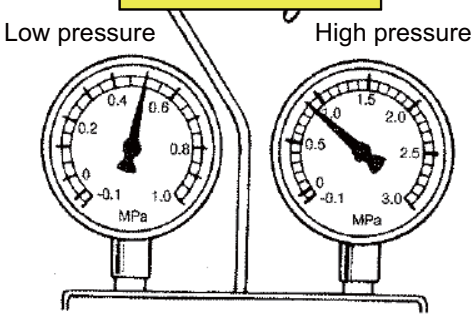
Water contamination in the cycle



● Condition
・Doesn't cool properly.
・After a certain amount of time, the low pressure side reading will gradually begin to indicate negative pressure.

● Remedy
・Replace the receiver dryer.
・Make sure to evacuate and drain out any water before charging with refrigerant.

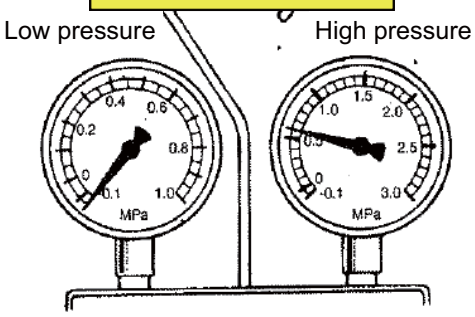
Faulty compression in the compressor



● Condition
・Doesn't cool properly (doesn't cool).
・The low pressure side reading is high while the high pressure side reading is low.

● Remedy
・Replace the compressor.
・Replace the receiver dryer.

Refrigerant does not circulate (clogged cycle)



● Condition
・Doesn't cool properly (doesn't cool).
・The low pressure side reading indicates negative pressure. If the cycle tends to get clogged, the gauge gradually begins to show a negative pressure.

● Remedy
・Replace the receiver dryer.
・Replace the expansion valve.
※In case of presence of soiling such as powdered metal, internal damage of compressor is a possibility. Replace.

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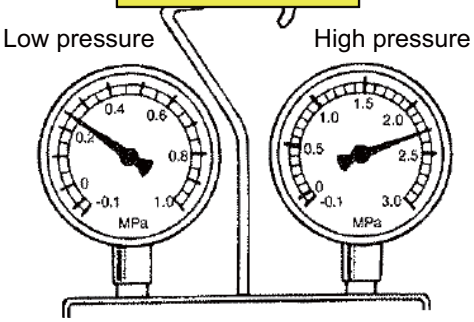
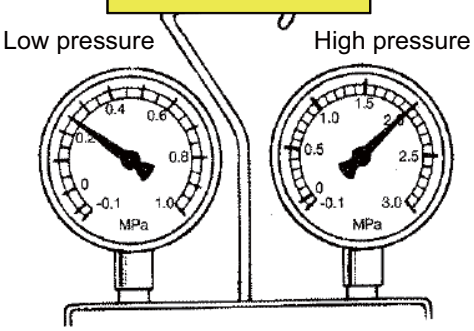
Air contamination in the cycle



● Condition
- Doesn't cool properly (doesn't cool).
- Pressure reading on both the low and high pressure sides are high.
- The low pressure pipe is not cold to the touch.

● Remedy
- Release the refrigerant and charge as needed.
- Make sure to evacuate properly.

Excessively opened expansion valve



● Condition
- Doesn't cool properly (doesn't cool).
- Pressure reading on both the low and high pressure sides are high.
- The low pressure pipe is covered with frost (condensation).

● Remedy
- Replace the expansion valve.